

VIITASAARI, Finland

WMO: 029150

Lat: 63.0822N

Lon: 25.8586E

Elev: 132

StdP: 99.75

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.5	-22.0	-28.3	0.3	-25.5	-24.5	0.4	-22.0	6.4	-3.2	5.8	-2.8	0.5	150	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	26.5	18.1	24.5	17.3	22.8	16.4	19.7	24.0	18.6	22.7	17.5	21.4	2.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	21.5	17.0	12.3	20.5	15.9	11.4	19.5	56.7	24.0	53.1	22.7	49.8	21.3	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.9	4.3	-27.3	28.6	4.8	2.3	-30.7	30.3	-33.6	31.6	-36.3	32.9	-39.8	34.6		
(5)			-27.3	20.9	4.7	1.8	-30.7	22.1	-33.5	23.2	-36.1	24.2	-39.6	25.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-8.0	-8.0	-3.5	2.9	9.1	14.3	17.4	15.6	10.3	3.9	-0.9	-4.7	(6)	
	DBStd	10.00	6.80	7.12	4.84	3.71	4.39	3.76	3.29	3.12	3.34	3.77	4.37	6.33	(7)	
	HDD10.0	2780	558	505	418	214	70	6	0	1	37	189	326	456	(8)	
	HDD18.3	5239	817	738	676	462	287	131	58	94	240	446	576	714	(9)	
	CDD10.0	631	0	0	0	1	42	134	229	176	47	2	0	0	(10)	
	CDD18.3	49	0	0	0	0	1	9	28	11	0	0	0	0	(11)	
	CDH23.3	302	0	0	0	0	16	62	176	48	0	0	0	0	(12)	
	CDH26.7	44	0	0	0	0	1	7	31	5	0	0	0	0	(13)	
(14) Wind	WSAvg	1.9	1.9	1.9	2.0	1.9	1.9	1.8	1.6	1.7	1.9	2.0	2.2	2.1	(14)	
(15) Precipitation	PrecAvg	600	39	34	31	31	45	67	81	66	53	57	49	45	(15)	
	PrecMax	759	84	65	62	63	118	126	147	118	134	112	111	76	(16)	
	PrecMin	422	8	2	4	10	12	24	19	21	18	12	5	12	(17)	
	PrecStd	76	17	18	13	15	25	28	34	27	30	26	23	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.9	8.7	17.6	25.7	27.9	29.5	27.2	20.7	13.0	9.3	6.1	(19)	
		MCWB	2.8	2.9	4.1	10.0	15.0	18.2	19.5	19.3	15.7	10.8	8.2	5.4	(20)	
	2%	DB	2.2	3.0	6.1	13.8	22.2	25.2	27.3	24.5	18.4	11.4	7.4	4.4	(21)	
		MCWB	1.5	1.9	2.8	7.9	13.8	17.0	19.1	17.6	14.3	10.3	6.6	3.7	(22)	
	5%	DB	1.2	1.4	4.2	11.3	19.2	22.9	25.2	22.8	16.7	10.5	5.9	3.2	(23)	
		MCWB	0.7	0.8	1.6	6.2	12.1	15.6	18.0	16.7	13.4	9.4	5.4	2.5	(24)	
	10%	DB	0.3	0.4	2.6	9.2	16.7	21.0	23.4	21.1	15.2	9.3	4.6	2.1	(25)	
		MCWB	-0.1	-0.1	0.7	5.1	10.7	14.8	17.3	16.1	12.7	8.3	4.0	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.2	4.8	11.1	16.6	20.0	21.8	20.2	16.3	11.7	8.5	5.6	(27)	
		MCDB	3.6	4.5	7.9	15.1	22.9	24.8	26.7	24.9	19.3	12.3	9.2	6.0	(28)	
	2%	WB	1.7	2.0	3.2	8.8	14.6	18.4	20.2	18.7	15.1	10.6	6.7	3.9	(29)	
		MCDB	2.1	2.8	5.4	13.0	19.9	22.9	24.7	22.9	17.4	11.3	7.2	4.4	(30)	
	5%	WB	0.8	0.9	2.1	6.8	13.1	16.9	19.2	17.7	14.1	9.6	5.4	2.6	(31)	
		MCDB	1.2	1.3	3.7	10.5	18.2	21.5	23.4	21.3	16.1	10.3	5.9	3.0	(32)	
	10%	WB	0.0	0.0	1.0	5.4	11.4	15.5	18.2	16.7	13.1	8.4	4.1	1.5	(33)	
		MCDB	0.3	0.4	2.2	8.7	15.9	19.9	22.2	20.0	14.8	9.2	4.6	2.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.9	7.4	8.2	8.9	8.2	7.8	7.3	6.0	4.0	3.3	4.4	(35)	
		MCDBR	4.4	4.7	8.5	11.9	12.9	11.1	10.4	9.7	7.8	4.9	4.3	3.9	(36)	
	5% WB	MCWBR	4.3	4.2	6.0	6.9	6.2	5.0	4.1	4.1	4.3	3.8	4.1	3.7	(37)	
		MCWBR	4.3	4.4	7.2	11.0	11.1	9.7	8.9	8.3	6.9	4.5	4.2	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.226	0.283	0.311	0.352	0.347	0.347	0.370	0.354	0.330	0.310	0.244	0.358	(40)	
		taud	2.012	2.131	2.204	2.334	2.454	2.479	2.429	2.465	2.520	2.398	2.102	2.641	(41)	
	Edn at Noon	Edn at Noon	396	642	781	829	870	876	840	823	776	629	370	230	(42)	
		Edh at Noon	32	65	90	100	96	96	99	87	68	52	29	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.18	0.73	2.23	3.67	4.82	5.25	4.97	3.80	2.16	0.84	0.21	0.08	(44)	
	RadStd	RadStd	0.02	0.16	0.24	0.42	0.54	0.41	0.51	0.42	0.25	0.16	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (39 neighbors)		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page